

Inelastic cross-sections of energetic protons in liquid water calculated by model dielectric functions and optical data

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Abstract

Total and differential inelastic inverse-mean-free-paths for the energy loss of protons in liquid water are calculated on the basis of the dielectric approach. Energetic protons, much above the Bragg peak, are considered where the first-Born-approximation is justified (~ 0.3 – 10 MeV). A model dielectric-response-function for the valence electrons of liquid water is constructed in accordance with the general properties of the Bethe-surface. A modified Drude-expansion model developed earlier provided the long-wavelength response, whereas several dispersion schemes were examined for introducing the momentum-transfer dependence. In particular, we have examined the δ -oscillator dispersion models of Ashley and Liljequist, and Ritchie's extended-Drude models. A comparison with optical dielectric models will also be presented for assessing the influence of the dispersion. In view of the limited experimental information for the inelastic properties of liquid water, the present work provides a comparative study of various – computationally tractable – schemes that may be used in analog Monte Carlo transport codes for protons and heavier ions in liquid water.

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1. Introduction

A comprehensive set of cross-section data for all the important interaction processes over the entire range of collision variables (impact energy, energy transfer, etc.) is a prerequisite for performing analog Monte-Carlo (MC) simulation of charged particle transport in matter. Proton inelastic models, much above the Bragg peak (i.e. electronic regime), are particularly important since

they may be extrapolated to heavier bare ions (of not too high z) by the simple z^2 -scaling suggested by the first Born approximation [1].

For ion beam applications in biomedicine, “condensed” (liquid/solid) water is the most relevant transport medium since it is the dominant component of soft tissue. However, the relevant experimental information is limited to either the optical limit or the region near the Bethe-ridge. In sharp contrast, direct measurements of inelastic cross-sections in water in the vapor phase are available, and, though incomplete, are often considered sufficient to support fully empirical models. This state of affairs has been responsible for the extensive use of vapor-based cross-section

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models which are then extrapolated linearly to unit density medium; the so-called gas-phase approximation [2]. In an effort to go beyond the gas-phase approximation and account for non-linear effects of the condensed state, the dielectric formalism has proved most useful. It is based upon the construction of a model dielectric-response function (DRF) for the medium of interest, which, in the framework of first-order perturbation theory (plane-wave Born approximation), codifies the deviation from the (classical) elastic scattering term, providing, in essence, the so-called inelastic form-factor [3]. The many-body aspects of the problem, however, render the formal evaluation of DRF problematic for all but the simplest systems. In view of the iterative process that underlines an MC simulation, it is desirable to look for simple – computationally tractable – models of the DRF, accounting, though, for the most important aspects of the theory. To that end, the so-called optical-data models, first applied by Ashley, implemented with appropriate dispersion schemes, appear to satisfy the above requirement [4]. Most importantly, this semi-empirical methodology allows for the inclusion of pertinent information regarding the condensed-state, such as electronic screening, correlation and collective effects. In the present work, total and differential inelastic inverse-mean-free-paths for proton-impact on liquid water are presented, calculated by a variety of dispersive and optical (i.e. non-dispersive) dielectric models for the valence shells, which represent the main inelastic channel in the energy range studied. It should be highlighted that all models examined start from the same optical-data model (developed earlier [5]) which is essential for a meaningful comparison of the various dispersion schemes.

2. Methodology

2.1. Born dielectric cross-sections

In the dielectric formalism, the (plane-wave) Born doubly differential inverse mean-free-path (IMFP) in energy loss E , and momentum transfer q expressed in terms of the DRF becomes [3]

$$\frac{d^2\Sigma(T, E, q)}{dE dq} = \frac{1}{\pi\alpha_0 Tq} \text{Im} \left[\frac{-1}{\varepsilon(E, q)} \right] \Theta[q - q_-(E, \tau)] \times \Theta[q_+(E, \tau) - q] \Theta[\tau - E], \quad (1)$$

where α_0 is the Bohr radius, τ is the proton kinetic energy, $T = (m/M)\tau$ is the kinetic energy of an electron with the same velocity to that of the proton (m and M are the electron and proton rest mass, respectively), ε is the complex DRF of the material ($\varepsilon = \varepsilon_1 + i\varepsilon_2$), $\text{Im}(\cdot)$ denotes the imaginary part of the argument, Θ is the unit step-function, and q_- , q_+ are the kinematically allowed minimum and maximum, respectively, momentum transfer in the collision. In the non-relativistic limit they are given by

$$q_{\pm} = \sqrt{2M}(\sqrt{\tau} \pm \sqrt{\tau - E}). \quad (2)$$

Note that, following the customary terminology for condensed targets, we make use of macroscopic cross-sections throughout; Σ is in units of 1/length. The use of q as scalar (instead of vector) in Eq. (1) implicitly assumes that the target is homogenous and isotropic, which, considering other uncertainties, is a reasonable assumption to make in the present context. The $\text{Im}[-1/\varepsilon(E, q)] = \varepsilon_2/(\varepsilon_1^2 + \varepsilon_2^2)$ term is called the energy-loss-function (ELF). It follows that, within the first Born approximation, inelastic cross-section calculations (and their moments) reduce to the evaluation of the ELF for all allowed values of E and q . The latter represents the so-called Bethe-surface of the material. Once the Bethe-surface has been modeled, it is straightforward to compute from Eq. (1) the energy-differential IMFP (designated here as DIMFP) and the integrated IMFP from the following expressions:

$$\frac{d\Sigma(T, E)}{dE} = \int \frac{d^2\Sigma(T, E, q)}{dE dq} dq, \quad (3)$$

$$\Sigma(T) = \int dE \int \frac{d^2\Sigma(T, E, q)}{dE dq} dq. \quad (4)$$

A formal mapping of the Bethe-surface, however, would require solving the many-body problem for excited states; a still impractical task considering typical computer resources. It is therefore desirable that semi-theoretical models be devised where

full advantage is taken of any existing experimental data.

2.2. The optical-data model

An optical-data model refers to an analytic representation of the dipole region ($q = 0$) of the Bethe-surface on the basis of experimental data at $q \approx 0$. In the present formulation, it represents the quantity $\text{Im}[-1/\varepsilon(E, q = 0)]$ providing the dependence of the inelastic probability on energy loss in the long wavelength limit. The use of optical-data models was first suggested by Ashley [4] and has been widely employed thereafter (e.g. [6] and reference therein). It offers distinctive advantages, such as the inclusion of the characteristic optical spectrum of the material accounting for electronic screening, correlation and (possible) collective effects. The data needed may be obtained by either electron energy-loss spectroscopy at forward angles, or optical (VUV and soft X-rays) absorption spectra. A detailed description of our optical model has been given elsewhere [5]; here, we will merely outline its main aspects. For experimental data we use the reflectance measurements of the Oak Ridge group on liquid water covering the 7–25 eV energy range [7], whereas for an analytic model we use the following long-wavelength Drude dielectric function:

$$\varepsilon(E, 0) = 1 + f_j \frac{E_p^2}{E_j^2 - E^2 - i\gamma_j E}, \quad (5)$$

where $E_p = \hbar(4\pi NZe^2/m)^{1/2}$ is the nominal plasma energy of the system with N the density of molecules in the liquid phase and Z their atomic number and f_j the fraction of electrons with characteristic oscillator energy and width $E_j = \hbar\omega_j$ and $\gamma_j = \hbar g_j$, respectively (ω and g represent the angular frequency and damping coefficient, respectively). For each transition mode (j), the model parameters $\{f_j, \gamma_j, E_j\}$ were deduced by fitting the experimental data on $\varepsilon_2(E, 0) \equiv \text{Im}[\varepsilon(E, 0)]$ using a linear superposition of Drude functions derived from Eq. (5). Following a suggestion by Ritchie [8], we represent the discrete by “derivative” Drude functions, which are more sharply peaked, whereas we keep the “normal” Drude functions for the continuum. The analytic representation of the valence

transitions of the liquid was guided by a parametrization first established for the free molecule appropriately adjusted, so that sum-rules are satisfied to within 1–2%. The various parameters of the optical-data model along with a more detailed discussion may be found in [5].

2.3. Dispersion models

A dispersion algorithm provides the momentum-transfer dependence of the energy loss spectrum established by the optical-data model. There is no definite method for extending the $\text{Im}[-1/\varepsilon(E, q = 0)]$ to $q > 0$ for realistic materials. Therefore, a heuristic approach is sought for, where one tries to account for the most important features of the Bethe-surface of the material, such as, for example, the development of the Bethe-ridge and the free-electron limit. For free-electron-like materials the use of Lindhard’s (or, better, Mermin’s) dielectric function [9], or simplifications thereof [6], directly provides the extension to the q -space. Despite the possibility of applying such functions to the valence band of some non-conducting solids with small band gap, their application to organic materials is questionable. In the present study the δ -oscillator schemes of Ashley and Liljequist, and Ritchie’s extended-Drude schemes are implemented to our optical-data model. To assess the influence of dispersion, comparisons will also be made with two non-dispersive models, which will be named here the optical-approximation and Bethe’s optical-term, respectively.

2.3.1. δ -oscillator dispersion schemes

The basis of the δ -oscillator dispersion algorithms is the fact that the optical-oscillator-strength density of an atom may be represented by the following sum [6]:

$$\frac{df(E, 0)}{dE} = \sum_j f_j \delta(E - E_j), \quad (6)$$

where f_j is the oscillator-strength for the particular transition j with resonance energy E_j . The δ -function in Eq. (6) may naturally be called an “optical”-oscillator (or δ -oscillator) [6]. This idea

may be used to extend the optical function to $q > 0$ by substituting E_j with a dispersion function $G_j(q)$ such as $G_j(0) = E_j$. By appropriately generalizing to the entire energy-loss spectrum and using the relationship between the oscillator-strength and the loss-function, $df(E)/dE = (2ZE/\pi E_p^2) \text{Im}[-1/\varepsilon(E)]$, the following is obtained:

$$E \text{Im} \left[\frac{-1}{\varepsilon(E, q)} \right] = \int_0^\infty E' \text{Im} \left[\frac{-1}{\varepsilon(E', 0)} \right] \times \delta(E - G(E'; q)) dE'. \quad (7)$$

In accordance with the previous discussion, the δ -function in Eq. (7) may now be called the δ -oscillator. Eq. (7) indicates that the $q > 0$ region of the Bethe-surface may be derived from its dipole limit ($q = 0$) by using δ -oscillators with appropriate dispersion functions $G(E'; q)$. In particular, Ashley has provided the following dispersion to be used in the δ -oscillator of Eq. (7) [10]:

$$G(E'; q) = E' + q^2/2m. \quad (8)$$

In the low- q domain, the above expression resembles the plasmon dispersion function of the free-electron gas, while asymptotically leading to the Bethe-ridge at high q -values. By inserting Eq. (8) into Eq. (7), the full ELF within Ashley's δ -oscillator scheme is obtained:

$$\text{Im} \left[\frac{-1}{\varepsilon(E, q)} \right] = \int_0^\infty E' \text{Im} \left[\frac{-1}{\varepsilon(E', 0)} \right] \times \frac{\delta(E - (E' + q^2/2m))}{E} dE'. \quad (9)$$

Ashley's dispersion scheme essentially amounts to a simple quadratic extension into the E - q plane. A drawback of the model is that discrete and continuum transitions are treated alike.

Liljequist has suggested an alternative dispersion scheme [11] based on Bohr's distinction between resonant-like and binary-like collisions. The former correspond to the dipole region of the Bethe-surface ($q = 0$) whereas the latter define the Bethe-ridge. On this basis, the following dispersion scheme was suggested [11]:

$$\begin{aligned} G(E'; q) &= E' \quad \text{for } q^2/2m \leq E', \\ G(E'; q) &= q^2/2m \quad \text{for } q^2/2m > E'. \end{aligned} \quad (10)$$

By inserting Eq. (10) in Eq. (7) the full ELF within Liljequist's δ -oscillator scheme is obtained:

$$\begin{aligned} \text{Im} \left[\frac{-1}{\varepsilon(E, q)} \right] &= \int_0^\infty E' \text{Im} \left[\frac{-1}{\varepsilon(E', 0)} \right] \frac{\delta(E - E')}{E} \\ &\quad \times \Theta(E' - q^2/2m) dE' \\ &\quad + \int_0^\infty E' \text{Im} \left[\frac{-1}{\varepsilon(E', 0)} \right] \\ &\quad \times \frac{\delta(E - q^2/2m)}{E} \Theta(q^2/2m - E') dE'. \end{aligned} \quad (11)$$

Eq. (11) may be simplified to

$$\begin{aligned} \text{Im} \left[\frac{-1}{\varepsilon(E, q)} \right] &= \text{Im} \left[\frac{-1}{\varepsilon(E, 0)} \right] \Theta(E - q^2/2m) \\ &\quad + E_p^2 \frac{\pi}{2} \frac{Z_{\text{eff}}(q)}{Z} \frac{\delta(E - q^2/2m)}{E}, \end{aligned} \quad (12)$$

where $Z_{\text{eff}}(q)$ is the effective number of target electrons which participate in collisions with energy transfer up to $q^2/2m$ as obtained from the relevant sum-rule. Eq. (12) indicates that, within Liljequist's scheme, the optical-ELF is used for the region of the q -plane extending up to $q^2/2m < E$, while the Bethe-ridge is developed with zero width at $q^2/2m = E$. The latter is weighted by the portion of the oscillator-strength contributing to this regime.

2.3.2. Extended-Drude schemes

Ritchie first suggested [12] that, once the optical spectrum has been represented by a Drude-type model, the q -dependence may be incorporated in the (empirical) model parameters by a substitution of the type $\{f_j, \gamma_j, E_j\} \rightarrow \{f_j(q), \gamma_j(q), E_j(q)\}$, as long as care is taken to satisfy the general sum-rules (for all q) and be able to reconstruct the Bethe-ridge for large q . Probably the simplest dispersion scheme in this context is the use of the triad $\{f_j, \gamma_j, E_j(q)\}$, where $E_j(q)$ is given by the impulse-approximation

$$E_j(q) = E_j(0) + q^2/2m. \quad (13)$$

In a more elaborate scheme, and in order to avoid the (unrealistic) free-electron-like dispersion of the discrete spectrum implied by Eq. (13), Ritchie and co-workers [8] suggested the restriction of Eq. (13)

to the continuum, i.e. $E_j(q) \rightarrow E_{j,\text{cont.}}(q)$, and the adoption of an empirically derived analytic GOS function established for the free-molecule, i.e. the use of the triad $\{f_j(q), \gamma_j, E_j(q)\}$. An analytic function of the general form

$$f_{\text{exc}}^{(j)}(q) = f_{\text{exc}}^{(j)}(0) \sum_n a_n^{(j)} q^n \exp(-b_n^{(j)} q) \quad (14)$$

has been suggested by Green and Miller [13] for the discrete spectrum. The empirical parameters (a_n, b_n) for the water molecule have been derived by Olivero et al. [14]. By means of the f -sum rule the q -dependence of the GOS for the continuum may be directly obtained from

$$f_{\text{ioniz}}^{(j)}(q) = f_{\text{ioniz}}^{(j)}(0) \frac{1 - \sum_j f_{\text{exc}}^{(j)}(q)}{1 - \sum_j f_{\text{exc}}^{(j)}(0)}, \quad (15)$$

assuming that the f -sum-rule is normalized to 1 (otherwise it should be replaced by the number of electrons per molecule, here 10). The GOS of the liquid will, obviously, bear a q -dependence that will be different from the one of the free molecule. The motivation, however, behind the adoption of such an approximation is the need to provide a reasonable variation with q beyond the impulse-approximation of Eq. (13). It is anticipated that the approximation made will introduce little uncertainty in the final calculations as long as the optical spectrum of the liquid has been used.

2.4. Non-dispersive models

It is instructive to compare the predictions of the various dispersive dielectric models presented above with the outcome of simpler models that will bypass the requirement of choosing a particular dispersion scheme. The crudest approximation would be to use the optical-ELF, instead of the $\text{Im}[-1/\varepsilon(E, q)]$, in Eq. (1) and calculate Eqs. (3) and (4) by performing the integration over $\ln(q)$. The extension of the dipole region, unchanged, to the entire E - q plane will be called here the optical-approximation. Using the limits $q_{\min} = (\hbar/\alpha_0)(E^2/4TR)^{1/2}$ and $q_{\max} = (\hbar/\alpha_0)(4T/R)^{1/2}$, the following DIMFP is obtained:

$$\frac{d\Sigma}{dE} = \frac{1}{\pi\alpha_0 T} \text{Im} \left[\frac{-1}{\varepsilon(E, 0)} \right] \ln(4T/E). \quad (16)$$

The theory of Bethe, on the other hand, provides a more sophisticated alternative to the above approximation by expanding the inelastic cross-section in inverse powers of T . Following Inokuti [15] we may write Bethe's asymptotic expansion of the DIMFP in the following convenient form:

$$\frac{d\Sigma}{dE} = A(E)T^{-1} \ln(T/R) + B(E)T^{-1} + O(E/T), \quad (17)$$

where $A(E) = (1/2\pi\alpha_0) \text{Im}[-1/\varepsilon(E, 0)]$, $B(E) = A(E) \ln[4(R/E)^2(K'\alpha_0)^2]$ and $O(E/T)$ denotes higher-order terms. It turns out that the value of the wave-number K' , as well as the next higher-order term, depend on the q -dependence of ELF. However, for sufficiently high T , where the Born approximation is formally valid, the first term in Eq. (16), being proportional to $T^{-1} \ln(T)$, dominates the expansion. Most importantly, it depends on the optical-ELF alone. Under this condition we are justified to truncate Eq. (16) to the first term and write

$$\frac{d\Sigma}{dE} = \frac{1}{2\pi\alpha_0} \text{Im} \left[\frac{-1}{\varepsilon(E, 0)} \right] T^{-1} \ln(T/R). \quad (18)$$

We will name Eq. (18) Bethe's optical-term.

3. Results

Calculated DIMFPs for 1 MeV protons in liquid water are presented in Fig. 1. Results have been obtained based on our Drude optical-data model extended to the E - q plane by the various dispersion schemes presented. In particular, from upper to lower peaks, we depict calculations based on Liljequist's δ -oscillator, Bethe's optical-term, Ritchie's extended-Drude models (the impulse approximation with and without the addition of an empirical GOS) and Ashley's δ -oscillator, respectively. Note that, for the particular range presented, only valence-shell electrons contribute to the inelastic process. Except from Bethe's term, the discrepancies are pronounced below about 50 eV energy loss (i.e. at the soft collision regime) due to the fact that all the dispersion schemes examined account for the Bethe-ridge at sufficiently high q -values (and, correspondingly, E -values).

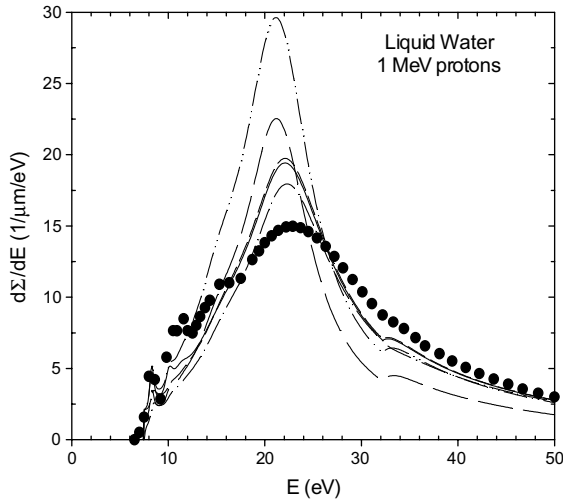


Fig. 1. Energy-transfer differential inverse mean-free-path, $d\Sigma/dE$, versus energy transfer, E , for 1 MeV protons in liquid water. The various curves represent calculations based on the same optical-data model (see text) extended to the energy-momentum plane by the following dispersion models; full-line: Ritchie's extended-Drude scheme (impulse approximation) with an empirical generalized-oscillator-strength for the free-molecule; short-dash line: Ritchie's extended-Drude scheme (impulse approximation); dash-line: Bethe's optical-term; dash-dot line: Ashley's δ -oscillator; dash-double-dot line: Liljequist's δ -oscillator; circles: calculations by Dingfelder et al. [16].

However, at the important range in the vicinity of the most probable energy loss (~ 20 eV), discrepancies by up to a factor of 1.7 are observed. Interestingly, the two Ritchie schemes are in almost perfect agreement for most part of the spectrum where the effect of the impulse approximation is dominant; they differ only at the lowest range where discrete transitions become pronounced. Results based on Ashley's δ -oscillator exhibit very similar trends with Ritchie's impulse scheme. This is explained by the quadratic extension that is characteristic of both schemes. Results from Bethe's optical-term are comparable with the other, more elaborate, schemes, e.g. only 15% higher peak values than Ritchie's schemes. However, it becomes gradually less trustworthy above the peak due to the neglect of optically forbidden transitions. This might be problematic when used for sampling the energetic secondaries in analog MC codes. Also included in Fig. 1 are the calculations by Dingfelder et al. [16]. The influence of

the optical-data model on the DIMFP becomes apparent when one compares the present results for Ritchie's scheme (with the inclusion of GOS) with those of Dingfelder and co-workers, who also used exactly the same dispersion scheme in their calculation. Despite the use of global constraints in both studies (sum-rules, I -value), and as expected from theory, differences in the analytic representation of the optical spectrum (not shown here, but see [5]) propagate in the resulting soft collision spectrum.

Secondary electron spectra for 1 MeV protons in liquid water are depicted in Fig. 2. In addition to the models depicted in Fig. 1, we also present calculations based on the optical-approximation (which was omitted from the linear scales of Fig. 1 for clarity purposes). The explanation of the curves follows that of Fig. 1. The contribution of the K-shell electrons to the continuum was modeled by the binary-encounter-approximation [1]. Their minimal contribution in the energy range of interest here justifies their representation by a semi-classical approximation. Fig. 2 shows that the optical-approximation results in peak values which are higher by up to a factor of 3 revealing the influence of the q -dependence of the ELF. The wrong asymptotic behavior of the non-dispersive

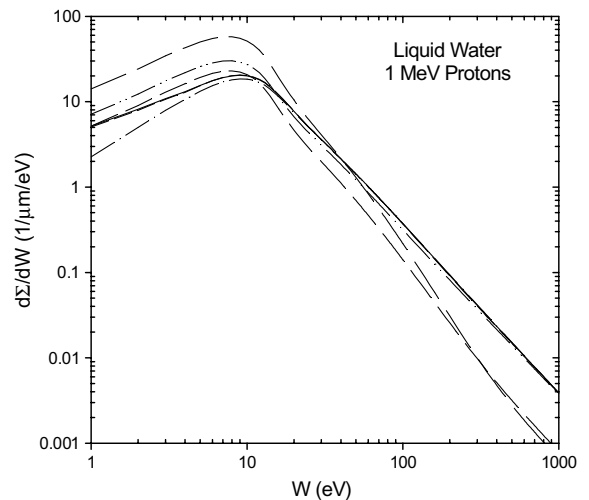


Fig. 2. Secondary electron spectra for 1 MeV protons in liquid water. Explanation of curves as in Fig. 1. The optical-approximation is represented by the long-dash line.

models is due to the Bethe-ridge, not accounted for in these models.

Platzman plots for 1 MeV protons in liquid water are presented in Fig. 3(a) and (b). The scaling in terms of E_1/R (see Fig. 3(a)) emphasizes large energy losses and should asymptotically reach the total number of electrons in the molecule (here 10). In contrast, the scaling in terms of R/E_1 (see Fig. 3(b)) emphasizes small energy losses and should follow the shape of the OOS multiplied by E . In the linear scale of the Platzman plots, the wrong asymptotic behavior of the non-dispersive models

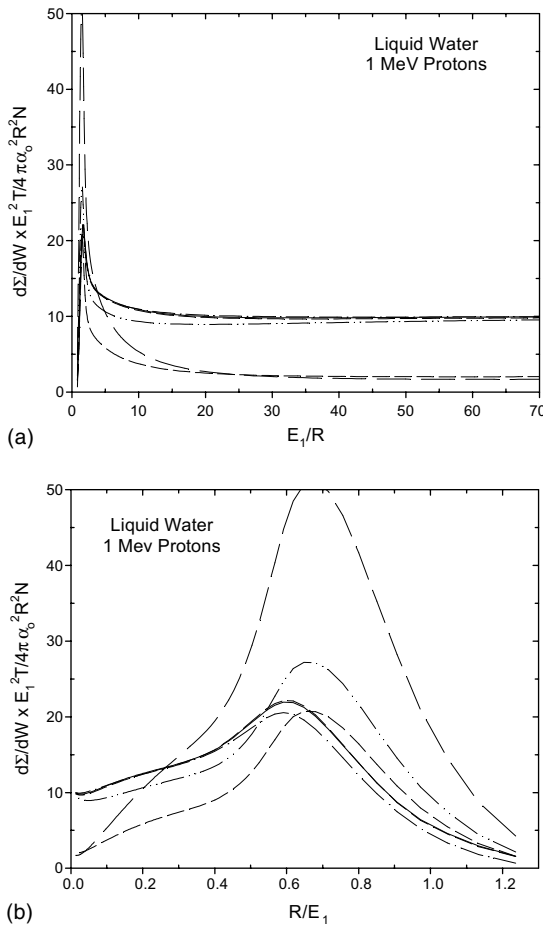


Fig. 3. (a,b) Platzman plots of liquid water using the model calculations depicted in Fig. 2. The two representations emphasize (a) large values of energy-transfer (E_1/R) and (b) small values of energy-transfer (R/E_1). Explanation of curves as in Figs. 1 and 2 (note: $E_1 \equiv W + B_1$).

(see particularly Fig. 3(a)) is most clearly seen. Inspection of Fig. 3(b) also reveals large differences between the various models in the soft collision regime, i.e. when collisions with small q -values are involved. As discussed earlier, this is the region where discrepancies between the predictions of the present dispersion models are expected to be larger.

Proton IMFPs in liquid water calculated by the various dielectric models examined are presented in Fig. 4. Differences resulting from the selection of the dispersion scheme are particularly noticeable below a few MeV proton impact reaching a factor of 1.6 at 0.3 MeV. Bethe's optical-term is in fair agreement with the more elaborate schemes exhibiting, at most, a 30% underestimation of Ritchie's model predictions. This might be of importance in transport simulations since Bethe's optical-term bypasses the integration over the q -variable and, therefore, is much less time-intensive. (Note, though, the comment with respect to its differential distribution.) In contrast, the optical-approximation is higher by a factor of 2 at 0.3 MeV. Despite the differences in the energy-loss spectra observed in Fig. 1, the IMFP calculation results of Dingfelder and co-workers are in very good agreement with current calculations (using Ritchie's scheme) due to the integration over the energy-transfer (and the sum-rules). Finally, a

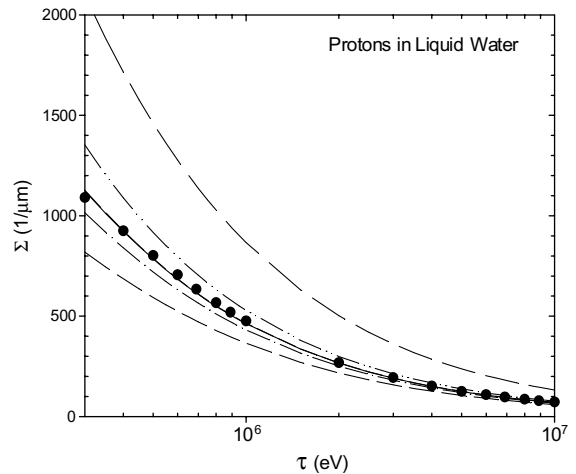


Fig. 4. Inelastic inverse mean-free-path, Σ , versus incident kinetic energy, τ , for protons in liquid water. Explanation of curves as in Figs. 1 and 2.

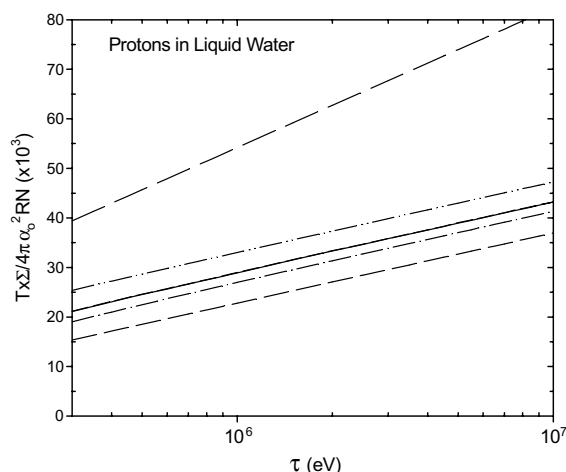


Fig. 5. Fano plot of liquid water using the model calculations depicted in Fig. 4. Explanation of curves as in Figs. 1 and 2.

Fano plot for proton impact in liquid water is presented in Fig. 5 for all models examined. Consistent with Bethe's theory, a straight line is obtained in the energy range studied. Dispersive models will deviate from linearity at lower energies due to non-dipole transitions, which correspond to higher-order terms in Bethe's expansion. The first Born approximation, however, progressively fails for these terms. Important information regarding the inelastic properties of the target may be obtained from the slope of the straight-line portion which is directly related to the dipole-matrix-elements (squared) for the particular transition (divided by 2 in the case of the optical-approximation), whereas the intercept with the vertical axis for the dispersive models would give the integral of the $B(E)$ term of Eq. (17).

4. Conclusion

We have presented a comparative analysis of various methodologies for calculating total and differential inelastic inverse-mean-free-paths for energetic protons in liquid water. The work follows the dielectric approach which has been found most suitable for particle transport simulation in condensed matter. Importantly, given the availability of optical data for a variety of organic and

tissue-equivalent materials, the models presented may be applied to other media of biological relevance. It was shown that whereas no single model may be uniformly accurate over the entire energy range of interest, the various dispersion schemes appear to differ by about a factor of 1.5 even at low energies which are largely responsible for phase effects and, thus, most problematic. In contrast, neglect of dispersion resulted in about a 5-fold underestimation of the hard collision spectrum. It should also be noted that the use of experimental optical data does have the distinct advantage that the influence of various many-body effects, such as long-range polarization, collective excitations and electron correlation, are already incorporated in the calculations. In view of the difficulties associated with modeling the production of low secondaries, which are decisive for track effects, the above realizations provides some degree of confidence with respect to the applicability of the present methodology.

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